

SQL Data Analysis Internship – Task 1

Student Details

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College Name	SSBT COET Bambori, Jalgaon
Branch	BE Computer Engineering
Year	BE
Submission Date	11/05/2026

Objective

The objective of this task is to create and analyze a Student Management Database using SQL. This project demonstrates database creation, table relationships, data insertion, data retrieval, joins, filtering, sorting, and aggregate functions for data analysis.

Database Creation

```
CREATE DATABASE StudentManagement;  
USE StudentManagement;
```

Students Table

```
CREATE TABLE Students (  
  student_id INT PRIMARY KEY,  
  student_name VARCHAR(100),  
  gender VARCHAR(10),  
  age INT,  
  city VARCHAR(50)  
);
```

Courses Table

```
CREATE TABLE Courses (  
  course_id INT PRIMARY KEY,  
  course_name VARCHAR(100),  
  instructor_name VARCHAR(100)  
);
```

Marks Table

```
CREATE TABLE Marks (  
  mark_id INT PRIMARY KEY,  
  student_id INT,  
  course_id INT,  
  marks INT,  
  exam_date DATE,  
  FOREIGN KEY(student_id) REFERENCES Students(student_id),  
  FOREIGN KEY(course_id) REFERENCES Courses(course_id)  
);
```

Important SQL Queries

1. SELECT * FROM Students;
 2. SELECT * FROM Students WHERE city='Pune';
 3. SELECT student_name, marks FROM Students s JOIN Marks m ON s.student_id=m.student_id WHERE marks > 80;
 4. SELECT AVG(marks) AS average_marks FROM Marks;
 5. SELECT MAX(marks) AS highest_marks FROM Marks;
 6. SELECT MIN(marks) AS lowest_marks FROM Marks;
 7. SELECT COUNT(*) AS total_students FROM Students;
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Key Concepts Used

- CREATE DATABASE
 - CREATE TABLE
 - PRIMARY KEY & FOREIGN KEY
 - INSERT INTO
 - SELECT Statements
 - WHERE Clause
 - JOIN Operations
 - GROUP BY & HAVING
 - Aggregate Functions
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Conclusion

The Student Management Database project was successfully implemented using SQL. Through this task, important SQL concepts such as table creation, joins, aggregation, filtering, and reporting were learned and applied practically.

Sample Student Performance Report

Student Name	Course	Marks
Aarav Sharma	SQL Basics	85
Sneha Patil	Data Analytics	88
Rahul Verma	Python	67
Priya Singh	SQL Basics	95

Final Result: The SQL Student Management Database task was completed successfully with proper database structure and detailed SQL analysis.